Annex-IV to Item No. 3

BS Computer Science)
(Robotics and Automation)

Semester-I

CL	1001	Introduction to ICT	0+1	
CS	1002	Programming Fundamentals	3+1	
NS	1001	Applied Physics	3+0	
MT	1003	Calculus & Analytical Geometry	3+0	
SS	1003	Pakistan Studies	3+0	
SS	1004	English Composition & Comprehension	2+1	
		Total	14+3	

Semester-II

CS	1004	Object Oriented Programming	3+1	CS 1002
EE	1005	Digital Logic Design	3+1	
MT	1006	Differential Equations	3+0	MT 1003
SS	1002	Islamic Studies/Ethics	3+0	
SS	1008	Communication & Presentation Skills	2+1	
		Total	14+3	

Semester-III

CS	2001	Data Structures	3+1	CS 1004
EE	2003	Computer Organization & Assembly Language	3+1	EE 1005
CS	1005	Discrete Structures	3+0	
MT	1004	Linear Algebra	3+0	
SS/MG		University Elective – I	3+0	
		Total	15+2	

Semester-IV

CS	2005	Database Systems	3+1	CS 2001
CS	2006	Operating Systems	3+1	CS 2001
CS	2009	Design & Analysis of Algorithms	3+0	CS 2001
MT	2005	Probability & Statistics	3+0	
SS/MG		University Elective – II	3+0	
		Total	15+2	

Semester-V

CS	3005	Theory of Automata	3+0	
CS	3001	Computer Networks	3+1	CS 2001
CS	3004	Software Design and Analysis	3+0	CS 1004
AI	2002	Artificial Intelligence	3+1	
RB-XXX		Robotics Technology	3+0	
		Total	15+2	

Semester-VI

CS	3009	Software Engineering	3+0	CS 3004
CS	3006	Parallel & Distributed Computing	3+0	CS 2006
SS	2007	Technical & Business Writing	3+0	SS 1008
CS	2008	Numerical Computing	3+0	
RB-XXX		Machine Learning for Robotics	3+0	CS 2001
			15+0	

Semester-VII

CS	4091	Final Year Project-I	0+3	Note 1
CS	3002	Information Security	3+0	
CS	4001	Professional Practices	3+0	
RB-XXX		Embedded Control for Robotics	3+0	EE 2003
MT		CS Supporting Elective	3+0	
			12+3	

Semester-VIII

CS	4092	Final Year Project-II	0+3	CS 4091
SS/MG		University Elective – III	3+0	
RB-XXX		Robotics Elective I	3+0	
RB-XXX		Robotics Elective II	3+0	
RB-XXX		Robotics Elective III	3+0	
		Total	12+3	

Robotics and Automation
Robotics Technology
Embedded Control for Robotics
Machine Learning for Robotics
Robotics in Healthcare
IoRT (Internet of Robotic Things)
Autonomous Systems
Cyber Security for Robotics
Robotics and Gaming
Perception Robotics
Robotic Interfacing
Robotic Actuators
Reinforcement Learning
Machine Learning



Electives
Stream Core

3 Stream Core
Any three Electives